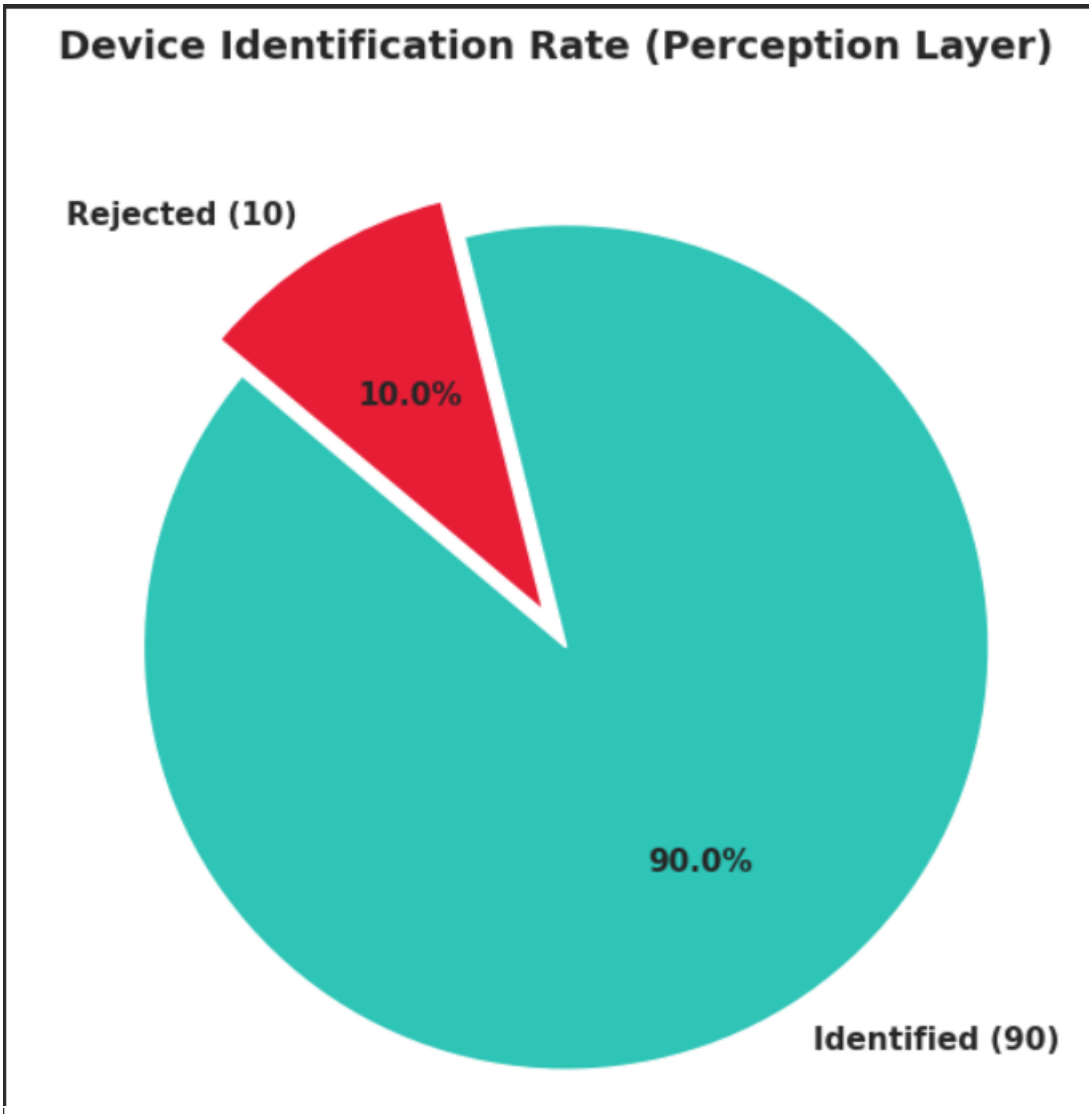
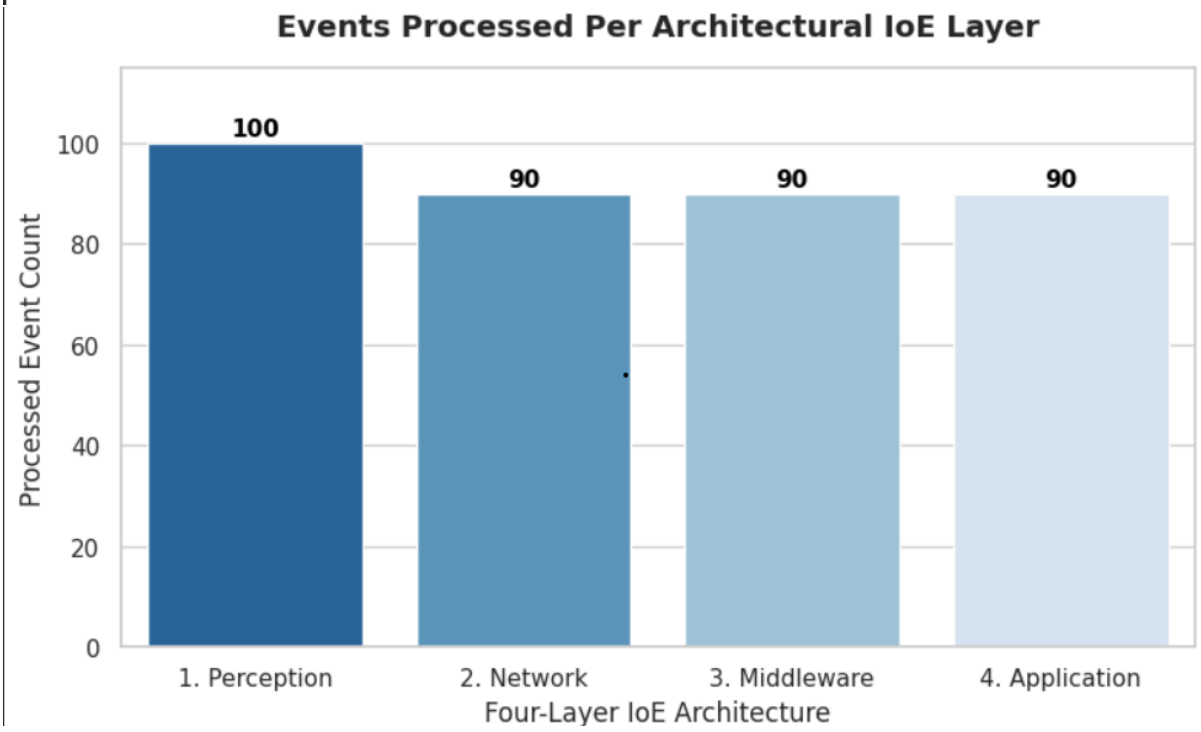
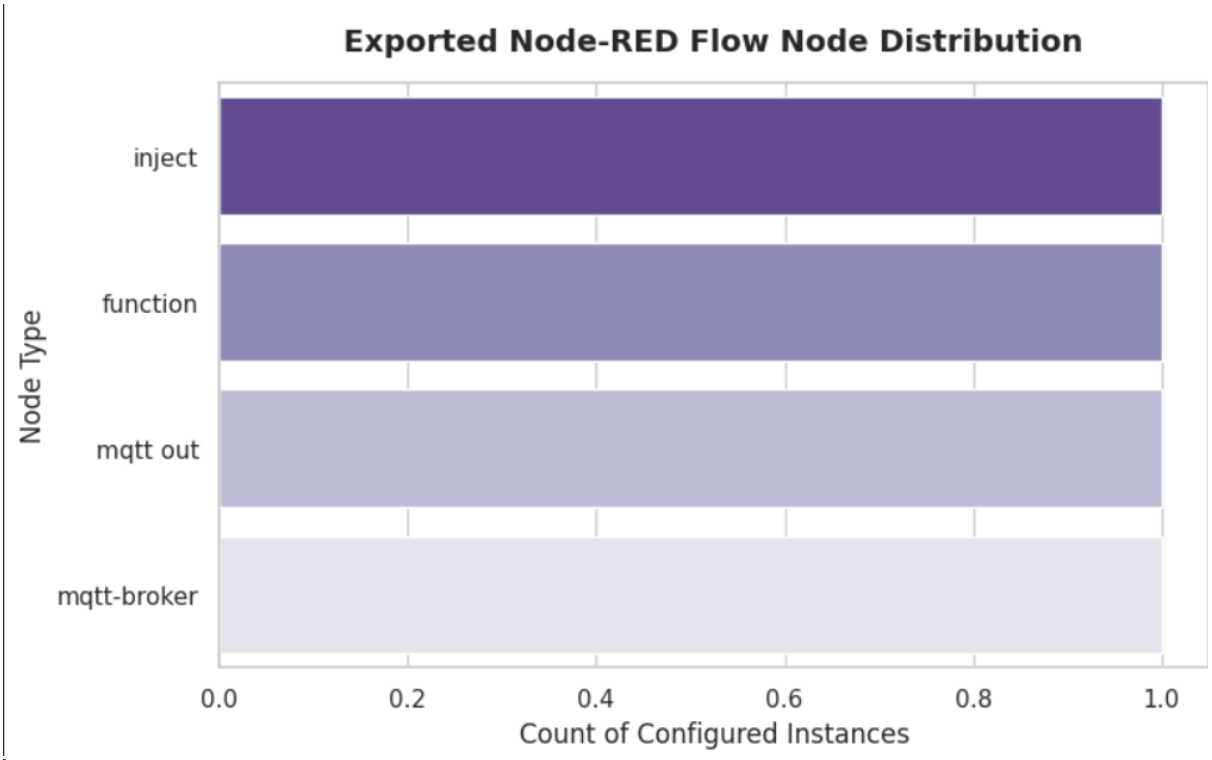
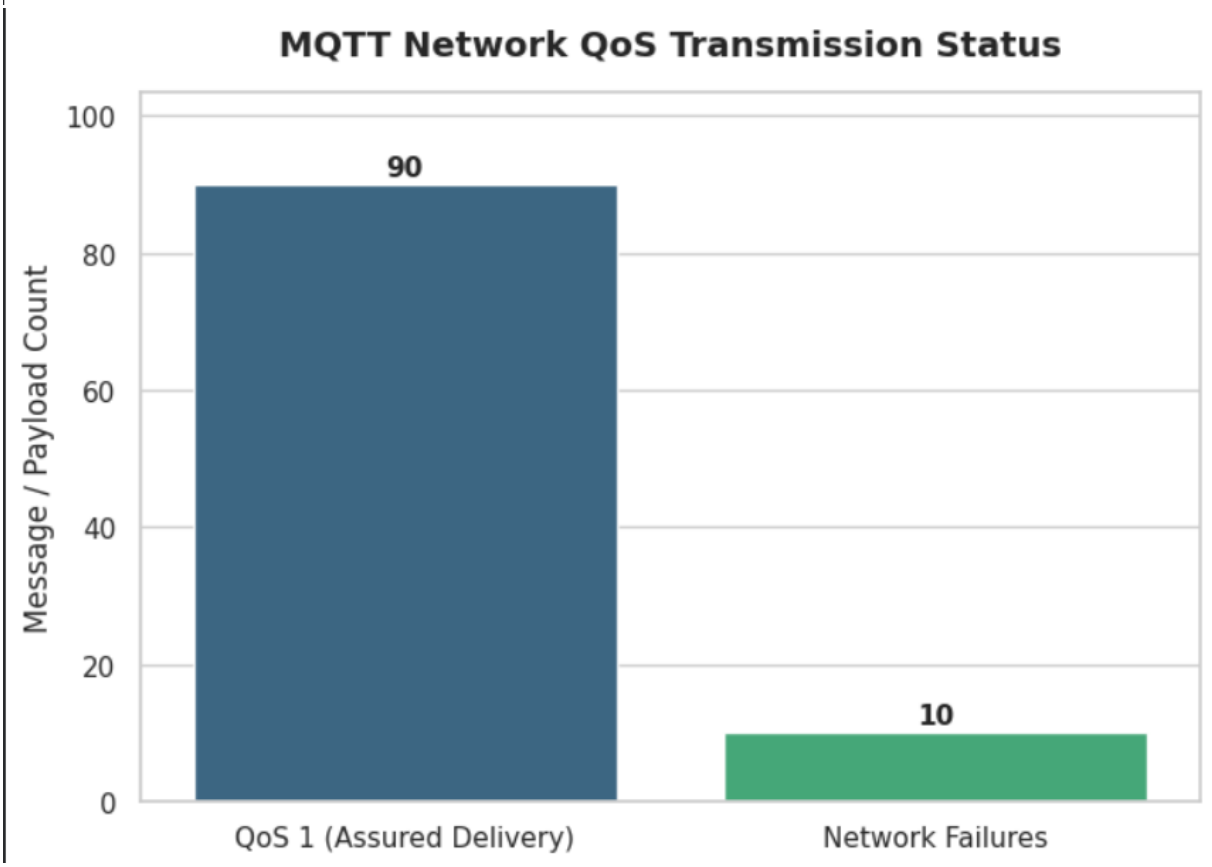
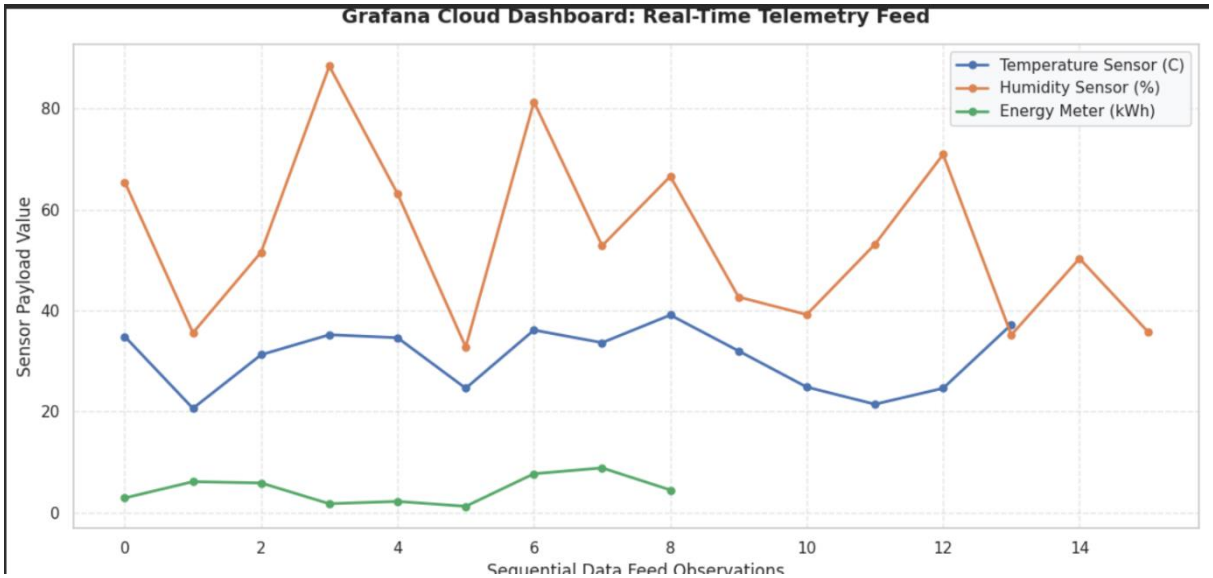
EXPERIMENT


Device Identification Rate (Perception Layer)
Rejected (10)
10.0%
90.0%
Identified (90)


Events Processed Per Architectural IoE Layer
Processed Event Count
0
20
40
60
80
100
100
90
90
90
1. Perception
2. Network
3. Middleware
4. Application
Four-Layer IoE Architecture


Exported Node-RED Flow Node Distribution
inject
function
mqtt out
mqtt-broker
Node Type
0.0
0.2
0.4
0.6
0.8
1.0
Count of Configured Instances


MQTT Network QoS Transmission Status
100
80
60
40
20
0
Message / Payload Count
90
10
QoS 1 (Assured Delivery)
Network Failures


Grafana Cloud Dashboard: Real-Time Telemetry Feed
Temperature Sensor (C)
Humidity Sensor (%)
Energy Meter (kWh)
Sensor Payload Value
0
20
40
60
80
0
2
4
6
8
10
12
14
Sequential Data Feed Observations